

Jones_ProjectBMC - Weak vs. Strong Examples

Project Context: Predictive Modeling for Bitcoin Volatility Forecasting: A Context-Aware Forecasting and Risk Insight System

1. Value Proposition

Weak Example

The app predicts Bitcoin volatility and helps users understand the market.

Strong Example

The system provides short-horizon Bitcoin volatility forecasts that help users interpret market instability more clearly than descriptive charts alone. Instead of acting as a trading bot or price-direction tool, it compares baseline and machine learning models under the same chronological framework, integrates contextual signals such as sentiment and uncertainty, and presents results through rankings, error summaries, and visual comparisons. Its value comes from disciplined forecasting, transparent comparison, and more useful risk insight.

2. Customer Segments

Weak Example

Crypto traders and investors who want to know what Bitcoin will do next.

Strong Example

Primary users include technically informed analysts, graduate researchers, and risk-focused cryptocurrency observers who need interpretable volatility insight rather than an automated trading interface. Secondary users include faculty reviewers, future developers, and research-oriented fintech teams interested in reproducible forecasting workflows. These users want a system that helps them compare forecasting approaches, find elevated-risk periods, and inspect whether contextual information improves predictive performance.

3. Channels

Weak Example

The product can be shared online and through a website.

Strong Example

Distribution channels include a dashboard-based web application, academic demonstrations, GitHub portfolio deployment, capstone presentations, and advisor review sessions. If expanded later, the system could also be delivered through API access, fintech analytics platforms, or institutional research environments. These channels fit the project because the application is analytical, digital, and best experienced through direct interaction with forecasts, rankings, and visual outputs.

4. Customer Relationships

Weak Example

Users will get instructions and be able to use the system.

Strong Example

The project depends on trust, interpretability, and clear communication. Users need validation feedback, transparent model comparisons, readable visualizations, and concise explanations of what each model is doing. The relationship is built around usability and responsible presentation rather than passive output delivery, and user feedback can guide refinement of dashboards, context variables, and explanation features.

5. Key Activities

Weak Example

Main activities include building the software, training models, and testing the app.

Strong Example

Key activities include Bitcoin data ingestion and validation, volatility feature engineering, contextual signal integration, baseline and machine learning model execution, chronological train-test evaluation, metric computation, dashboard construction, and export support. Additional activities include testing for leakage, maintaining reproducibility, and refining how results are presented to users. These activities matter because the system's credibility depends on methodologically sound forecasting and understandable outputs.

6. Key Resources

Weak Example

Important resources are data, developers, and software tools.

Strong Example

Critical resources include historical Bitcoin OHLCV data, contextual variables such as sentiment, investor attention, and uncertainty-sensitive indicators, Python-based data science and machine learning libraries, computing resources, dashboard frameworks, version control infrastructure, and technical documentation. These are just as important as methodological resources such as chronological validation procedures, evaluation metrics, and reproducibility controls that keep the forecasting pipeline defensible.

7. Key Partners

Weak Example

Partners could include data companies and software companies.

Strong Example

Key partners include cryptocurrency data providers, cloud or hosting services, open-source software communities, academic advisors, and research institutions that support financial analytics and machine learning work. Future partnerships could also involve fintech platforms or analytics vendors interested in integrating volatility insight tools. These partners matter because the project depends on reliable data, scalable infrastructure, and continued technical refinement.

8. Cost Structure

Weak Example

Costs include development, data, and testing.

Strong Example

Primary costs include data preparation, contextual feature collection and alignment, software development time, model training, evaluation, dashboard implementation, testing, maintenance, and possible hosting or storage expenses. If the project grows beyond the capstone stage, additional costs may come from premium data access, stronger deployment infrastructure, and ongoing retraining support. The main cost driver supports a forecasting system that is still reproducible, interpretable, and technically credible.

9. Revenue Streams / Sustainability Model

Weak Example

Users could pay for the app or the software in the future.

Strong Example

Potential sustainability models include subscription access to the forecasting dashboard, API licensing, institutional research use, and grant-supported development. In its current capstone form, the strongest sustainability argument is that the system serves as a research-backed prototype with academic, professional,

and portfolio value. Its long-term value comes from showing how volatility forecasting, contextual modeling, and user-facing analytics can be combined into one coherent system.

Why the stronger version is better

The stronger version defines what the system actually does, who it is for, how it creates value, and why each business model element fits the project. It avoids vague labels and keeps the focus on the project's real technical purpose: context-aware Bitcoin volatility forecasting, comparative model evaluation, and user-facing risk insight.